

Maximal Independent Set (MIS) Algorithm Performance Report

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1 Introduction

This report evaluates various implementations of the Maximal Independent Set (MIS) algorithm. MIS is a fundamental problem in graph theory with applications in distributed computing, wireless networks, and resource allocation. We compare a sequential implementation against parallel versions based on Luby's algorithm, including multi-core CPU and GPU implementations.

2 Graph Representation

The efficiency of graph algorithms is heavily influenced by how the graph is stored in memory. In this project, we implemented and compared two primary representations:

2.1 Standard Adjacency List

The `Graph` type is implemented as a `std::vector<std::vector<uint32_t>>`. While this representation is intuitive and allows for easy graph construction, it has several drawbacks for high-performance computing:

- **Memory Fragmentation:** Each vertex's adjacency list is a separate memory allocation, leading to scattered data across the heap.
- **Cache Misses:** Traversing the neighbors of multiple vertices often results in frequent cache misses due to the lack of data contiguity.
- **Overhead:** Each `std::vector` object carries internal metadata (pointers, size, capacity), which adds significant memory overhead for large graphs with millions of vertices.

2.2 Compressed Sparse Row (CSR)

To address the limitations of the adjacency list, we implemented the `GraphCSR` class. This format stores the entire graph structure using two contiguous arrays:

- **offsets:** An array of size $N + 1$ where `offsets[i]` marks the starting position of vertex i 's neighbors in the `edges` array.
- **edges:** A single array of size M (total number of edges) containing all neighbor indices stored back-to-back.

The CSR format offers several advantages:

- **Cache Locality:** Since all edges are stored contiguously, traversing neighbors is highly cache-efficient, significantly speeding up neighbor traversal phases.
- **Memory Efficiency:** CSR eliminates the metadata overhead of individual vectors and reduces memory fragmentation.
- **GPU Compatibility:** The contiguous nature of CSR makes it the standard format for graph processing on GPUs, as it allows for efficient coalesced memory accesses.

3 Algorithms

In this section, we describe the main algorithms implemented for the Maximal Independent Set (MIS) problem.

3.1 Sequential MIS

The sequential algorithm follows a greedy approach. It iterates through all vertices and adds a vertex to the MIS if none of its neighbors have been added yet. This algorithm has a linear time complexity of $O(N + M)$, where N is the number of vertices and M is the number of edges.

```
1 for (std::size_t i = 0; i < n; i++) {  
2     if (active[i]) {  
3         mis.push_back(i);  
4         for (auto &son : g[i])  
5             active[son] = 0;  
6     }  
7 }
```

3.2 Luby's Algorithm

Luby's algorithm is a randomized parallel algorithm for finding a Maximal Independent Set. It is designed to work in a synchronous, round-based manner, making it highly suitable for parallel architectures like multi-core CPUs (via OpenMP) and GPUs (via CUDA).

The algorithm proceeds in discrete rounds, where each round consists of the following steps:

1. **Priority Assignment:** Each currently active vertex v independently chooses a random priority $p(v)$ from a large uniform distribution.
2. **Local Maxima Selection:** A vertex v is added to the independent set if its priority $p(v)$ is strictly greater than the priorities of all its active neighbors $N(v)$. Ties are typically broken using vertex indices.
3. **Pruning:** Any vertex selected in the previous step, along with all of its neighbors, is marked as inactive and removed from further consideration.

The process repeats on the remaining induced subgraph until no active vertices remain.

Complexity: Theoretically, Luby's algorithm is very efficient. It has been proven that the algorithm finds a Maximal Independent Set in $O(\log n)$ rounds with high probability, where n is the number of vertices. Since each round involves traversing edges, the total work is proportional to the number of edges M , but the "depth" of the computation is logarithmic, allowing for significant speedups on parallel hardware.

3.2.1 OpenMP Parallelization

We utilized OpenMP to parallelize Luby's algorithm on multi-core CPUs, given these factors:

- **Shared Memory Model:** Luby's algorithm requires threads to frequently access the priorities and status of neighboring vertices. OpenMP's shared memory architecture allows all threads to access the global graph structure and priority arrays without the overhead of explicit message passing.
- **Data Parallelism:** The core operations of each round—priority assignment and local maxima selection—are naturally data-parallel. Using `#pragma omp parallel for` allowed us to distribute these tasks across available CPU cores with minimal code modification.
- **Dynamic Load Balancing:** In irregular graphs, such as Scale-Free networks, vertex degrees vary significantly. OpenMP's dynamic scheduling helps mitigate the "straggler" problem where threads processing high-degree nodes would otherwise bottleneck the entire round.
- **Synchronization Primitives:** OpenMP provides efficient primitives like `atomic` and `critical` sections, which were essential for managing shared state updates (e.g., adding nodes to the MIS or updating the "any active" flag) safely across threads.

3.3 Improved Luby’s Algorithm

The improved version refines the base parallel Luby algorithm by reducing synchronisation overhead, eliminating explicit random number generation, and exploiting locality more effectively.

The key optimisations are:

- **Active Node Tracking:** Instead of scanning the entire vertex set each round, a dedicated vector `active_nodes` stores only those vertices still eligible for the independent set. After each round, survivors are compacted into a fresh list for the next iteration.
- **Deterministic, On-the-Fly Priorities:** Each vertex’s random priority is computed from the current round number and its own index using a fast, `splitmix64` hash function. This avoids an expensive thread-local RNG pass, eliminates the need for a global priority array, and guarantees a reproducible pseudo-random order without additional memory traffic.
- **Unified Max-Detection and Deactivation:** The search for local maxima and the immediate deactivation of winners and their neighbours are fused into a single parallel region. This fusion halves the number of global barriers per round compared to the basic version, reducing overhead.
- **Dynamic Load Balancing:** The neighbour-checking loop uses `schedule(dynamic, 256)` to adapt to the highly irregular degree distributions of scale-free and other non-uniform graphs.
- **Thread-Local Aggregation with Critical Merge:** Both the newly selected independent set vertices and the surviving active nodes are collected into thread-local lists during their respective loops. A single critical section per thread then appends the local results to a global vector. This pattern is used in the unified max-detection/deactivation region (for MIS additions) and again in the lightweight survivor compaction region.

3.4 Luby GPU Implementation

The GPU version of Luby’s algorithm leverages the massive parallelism of CUDA architectures. The implementation is based on the CSR format and follows an iterative process where several kernels are launched per round:

- **Iterative Kernel Execution:** The algorithm operates in rounds. In each round, we launch a sequence of specialized CUDA kernels: one to identify candidates (local maxima) and another to update the MIS and prune neighbors. This cycle continues iteratively until all vertices have been processed.
- **Fused Kernels:** We combine priority assignment and candidate selection into a single kernel (`select_assign_priorities_fused_candidates_kernel`) to reduce global memory traffic and kernel launch overhead.
- **Restricted Pointers:** We utilize the `__restrict__` keyword for all input arrays (offsets, edges, active status). This informs the compiler that these pointers do not alias, enabling more aggressive load/store optimizations and better utilization of read-only data caches.
- **Deterministic Hashing:** Instead of generating and storing random numbers for all nodes in every round, we use a lightweight `hash_priority` function based on the vertex ID and iteration number. This reduces memory bandwidth requirements as priorities are computed on-the-fly.
- **Minimized Host-Device Transfers:** Only a single integer (`any_active`) is copied from device to host at the end of each round to determine convergence, keeping the hot loop mostly on the GPU.

4 Experimental Setup

To rigorously evaluate the performance of our implementations, we utilize two distinct random graph models that represent different topological characteristics commonly found in real-world applications.

4.1 Erdős-Rényi Model (Uniform Sparse)

The first model is the standard Erdős-Rényi random graph, specifically the $G(n, p)$ model [2]. In this model, each potential edge between any two vertices exists independently with a constant probability p . For our "Uniform Sparse" benchmarks, we set p such that the average number of edges is $M = c \cdot N$, with c typically set to 10.

This graph type serves as a baseline for "average" case performance. The degree distribution follows a Poisson distribution for large n , meaning most vertices have approximately the same number of neighbors. This relative uniformity simplifies workload balancing across parallel threads.

4.2 Barabási-Albert Model (Scale-Free)

The second model is the Barabási-Albert (BA) model [3], which generates scale-free graphs using a preferential attachment mechanism. The process starts with a small clique of m_0 nodes, and new nodes are added one by one, each connecting to m existing nodes with a probability proportional to their current degree.

Scale-free graphs are characterized by a power-law degree distribution, $P(k) \sim k^{-\gamma}$. These graphs contain "hubs"—a few vertices with extremely high degrees—while the vast majority of vertices have very few connections. This topology is representative of social networks and the World Wide Web. For MIS algorithms, scale-free graphs present a significant challenge for parallelization due to extreme workload imbalance, making optimizations like dynamic scheduling and CSR layout essential.

5 Results

In this section, we present the performance results for the various implementations on different graph sizes and densities. All benchmarks were performed using the Compressed Sparse Row (CSR) representation.

5.1 Scaling with Graph Size

We evaluated the algorithms on Uniform Sparse graphs ($c = 10$) of varying sizes, ranging from 10^5 to $5 \cdot 10^7$ nodes. The results are summarized in Table 1.

Table 1: Performance scaling with problem size N (Density $c = 10$, Uniform Sparse)

N	Sequential (ms)	Luby CSR (ms)	Luby Improved (ms)	Luby GPU (ms)
10^5	0.392	0.756	0.528	5.057
10^6	4.961	38.911	11.362	15.969
10^7	75.168	265.706	176.964	149.048
$5 \cdot 10^7$	707.244	2107.693	759.012	805.434

5.2 Scaling with Graph Density

We also investigated the impact of graph density on performance by varying the edge constant c for a fixed number of nodes $N = 5 \cdot 10^6$. The results are shown in Table 2.

Table 2: Performance scaling with graph density c (Nodes $N = 5 \cdot 10^6$, Uniform Sparse)

c (Edges/ N)	Sequential (ms)	Luby CSR (ms)	Luby Improved (ms)	Luby GPU (ms)
2	28.151	125.158	21.629	32.262
10	35.344	108.079	50.415	70.863
50	43.759	193.763	109.687	308.092
100	45.117	182.206	78.040	620.733

5.3 Performance on Scale-Free Graphs

We also evaluated the performance on Scale-Free graphs ($m = 5, m_0 = 10$) to observe how the algorithms handle highly unbalanced degree distributions. The results are summarized in Table 3.

Table 3: Performance scaling with number of nodes N (Scale-Free graphs, $m = 5$)

N	Sequential (ms)	Luby CSR (ms)	Luby Improved (ms)	Luby GPU (ms)
$5 \cdot 10^6$	55.530	124.323	53.586	78.322
10^7	115.266	225.880	176.630	150.709
$2 \cdot 10^7$	247.660	515.658	229.322	301.812

6 Weighted MIS

6.1 Problem Formulation

We recall that the standard Maximal Independent Set (MIS) is a subset of vertices $S \subset V$ of a graph $G = (V, E)$ such that no two vertices in S are joined by an edge, and no new vertex can be added to S without creating an edge. Now, we extend this framework by adding a weight $w_v \in \mathbb{R}^+$ for every vertex $v \in V$. In this setting, the goal is now to find a MIS with the maximum possible total weight **in practice**. It is important to note that finding the **maximum weight independent set** (i.e., the globally optimal solution) is NP-hard in general. In this view, we will focus on weight-based heuristic algorithms that find a valid MIS while favoring vertices with high weights, and then compare their solution quality empirically.

6.2 Algorithms

We decided to study the performance of the following two algorithms, and then benchmark their performance based on the initial weight assignment.

6.2.1 Algorithm 1: Weighted Greedy

A vertex v is added to the MIS if and only if it has the highest weight among its active neighbors:

$$v \text{ wins} \iff w_v \geq w_u \quad \forall u \in N(v), u \text{ active} \quad (1)$$

Ties are broken by vertex index. This approach is based on Basagni [1].

6.2.2 Algorithm 2: Weighted Sampling

Each active vertex v draws $U_v \sim \text{Uniform}(0, 1)$ and computes a priority:

$$p_v = U_v^{1/w_v} \quad (2)$$

Vertex v wins the round if $p_v > p_u$ for all active neighbors u . That is, we have that the vertices with higher weights have the higher priority.

6.3 Weight Distributions

Uniform: Each vertex v is assigned a weight drawn uniformly at random:

$$w_v \sim \text{Uniform}(1, 10) \quad (3)$$

Exponential: Each vertex v is assigned:

$$w_v = 1 + X, \quad X \sim \text{Exponential}(\lambda) \quad (4)$$

The $+1$ shift ensures $w_v > 0$, which is required by the power trick in Algorithm 2.

Clustered: A fraction $\rho = 0.1$ of vertices are randomly designated as hotspots:

$$w_v = \begin{cases} 10 & \text{with probability } \rho \\ 1 & \text{otherwise} \end{cases} \quad (5)$$

6.4 Experimental Results

The following results were captured from the weighted MIS benchmark suite ($N = 1,000,000$, 5 graphs, 10 runs each).

Algorithm	Graph Type	Mean Time	Graph Std	Avg Run Std	Mean Weight
Weighted Greedy	Uniform Sparse (uniform)	48.346 ms	10.319 ms	19.023 ms	221,205.58
Weighted Sampling	Uniform Sparse (uniform)	101.725 ms	62.736 ms	86.907 ms	169,862.79
Weighted Greedy	Scale-Free (uniform)	34.835 ms	5.984 ms	7.885 ms	2,220,850.60
Weighted Sampling	Scale-Free (uniform)	65.297 ms	47.481 ms	41.514 ms	1,998,436.76
Weighted Greedy	Uniform Sparse (exp)	44.856 ms	2.182 ms	7.825 ms	183,315.81
Weighted Sampling	Uniform Sparse (exp)	65.743 ms	33.578 ms	24.710 ms	102,926.19
Weighted Greedy	Scale-Free (exp)	53.372 ms	44.933 ms	35.131 ms	1,338,124.13
Weighted Sampling	Scale-Free (exp)	40.659 ms	5.625 ms	11.284 ms	1,133,123.09
Weighted Greedy	Uniform Sparse (clustered)	63.872 ms	4.375 ms	7.339 ms	163,565.00
Weighted Sampling	Uniform Sparse (clustered)	45.382 ms	3.298 ms	5.214 ms	107,985.08
Weighted Greedy	Scale-Free (clustered)	41.219 ms	12.018 ms	25.399 ms	1,051,920.60
Weighted Sampling	Scale-Free (clustered)	62.458 ms	25.564 ms	44.257 ms	877,655.86

6.5 Discussion

The results indicate that the Greedy approach consistently finds the MIS with higher total weight across all graph types and initial weight distributions. This can be explained by the fact that stochastic sampling can occasionally make the algorithm give a low-weight neighboring vertex a higher priority, thus permanently excluding the high-weight vertex from the MIS selection.

We also see that on Scale-Free graph types, the mean weight results are much closer than those on the Uniform-Sparse graph types (gap of 11–20% on Scale-Free vs. 30–78% on Uniform Sparse).

We see that the exponential and clustered initial weight distributions improve greedy’s advantage (78% and 52% gaps respectively on Uniform Sparse, vs. 30% for uniform weights). The reason for this is that in these settings, we have relatively fewer very high-weight vertices, and the greedy approach always captures them. Meanwhile, the sampling approach may exclude them in case a low-weight vertex is selected instead at random.

References

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